

SABIHA GOKCEN, Turkey

WMO: 170630

Lat: 40.898N

Lon: 29.303E

Elev: 325

StdP: 14.52

Time Zone: 3.00 (E03)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.6	31.6	21.1	16.0	35.1	23.4	17.8	36.5	27.4	36.9	24.8	40.9	8.4	30	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.9	90.0	70.0	87.9	69.9	86.0	69.8	74.8	82.6	73.8	81.6	72.6	80.4	12.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	122.3	78.3	71.4	117.5	77.8	69.9	111.4	77.1	38.5	82.9	37.5	82.3	36.5	80.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.1	20.8	18.7		DB	23.9	95.4	3.6	3.4	21.3	97.8	19.2	99.8	17.2	101.7	14.6	104.2
				WB	22.7	76.1	3.5	1.7	20.2	77.3	18.1	78.3	16.2	79.2	13.6	80.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	43.4	45.2	48.7	55.7	64.6	72.3	76.5	77.5	70.6	61.5	54.3	46.6	
	DBStd	13.35	7.40	7.42	6.83	6.47	5.76	4.68	3.50	3.46	4.71	5.61	6.65	6.90	
	HDD50	707	230	170	109	16	0	0	0	0	0	2	30	150	
	HDD65	3136	670	556	505	286	77	3	0	0	10	134	325	570	
	CDD50	4293	25	35	70	187	454	670	820	851	617	358	160	46	
	CDD65	1246	0	0	0	7	66	223	355	386	177	26	5	1	
	CDH74	9186	0	0	3	61	419	1557	2907	3149	988	90	11	1	
	CDH80	2818	0	0	0	4	84	446	971	1084	223	6	0	0	
Wind	WSAvg	8.8	8.4	8.9	8.8	8.4	8.4	8.9	10.2	10.6	9.1	8.6	7.4	7.8	
Precipitation	PrecAvg	25.2	3.8	2.4	2.4	1.9	1.1	0.9	0.7	1.1	1.4	2.5	3.1	4.4	
	PrecMax	34.3	8.8	5.9	6.1	3.9	3.5	2.5	2.9	5.3	6.5	7.0	7.5	10.5	
	PrecMin	16.9	0.4	0.4	0.3	0.3	0.1	0.0	0.0	0.0	0.0	0.3	1.1	1.2	
	PrecStd	5.1	1.9	1.2	1.5	1.0	0.7	0.7	0.7	1.3	1.5	1.6	1.5	2.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	66.8	71.7	80.3	86.2	91.8	93.6	93.2	89.7	80.7	75.5	69.5	
		MCWB	51.9	53.7	55.6	60.7	64.7	69.0	69.4	70.3	68.5	63.8	59.3	55.6	
	2%	DB	60.7	62.6	67.6	75.2	82.3	87.9	89.8	89.9	85.5	76.7	71.5	63.0	
		MCWB	50.4	51.8	53.1	58.1	63.7	68.9	70.0	70.8	68.0	62.7	57.9	53.4	
	5%	DB	57.5	59.2	64.0	71.3	78.8	84.5	87.6	87.9	82.3	73.4	68.0	60.6	
		MCWB	48.6	50.2	52.0	56.3	62.8	67.5	69.7	70.9	66.9	61.9	56.9	52.0	
	10%	DB	55.0	56.9	60.4	67.7	75.2	82.2	85.5	85.9	79.2	70.1	64.5	57.3	
		MCWB	47.7	49.1	50.5	55.4	61.4	67.1	69.6	71.1	65.9	61.2	56.0	50.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	55.4	56.4	61.8	68.0	74.2	75.8	76.6	73.3	67.1	62.1	56.6
			MCDB	61.9	64.5	68.2	76.5	79.2	83.9	85.8	83.5	82.5	74.8	70.0	64.9
2%		WB	50.9	52.8	54.3	59.3	65.6	72.1	74.2	75.5	71.3	65.2	59.6	54.8	
		MCDB	58.8	60.5	64.4	71.7	76.5	80.2	82.7	82.7	78.7	72.1	68.2	62.2	
5%		WB	49.5	50.8	52.6	57.6	64.1	70.6	72.9	74.5	69.9	63.8	57.8	52.8	
		MCDB	56.5	58.4	61.6	69.3	74.6	78.9	81.1	81.8	77.4	69.9	65.8	59.3	
10%		WB	47.9	49.4	51.0	55.6	62.8	69.4	71.9	73.4	68.3	62.5	56.3	50.9	
		MCDB	53.9	56.5	59.1	65.3	73.2	78.2	80.0	80.8	76.3	68.2	63.3	56.0	
Mean Daily Temperature Range	5% DB	MDBR	10.9	11.4	14.1	16.7	16.2	16.4	17.1	16.9	15.9	13.2	12.8	11.3	
		MCDBR	14.6	15.5	19.7	22.3	20.9	20.2	20.0	18.9	19.8	17.7	16.5	14.3	
		MCWBR	8.1	7.9	9.5	10.0	8.3	7.3	6.8	5.9	7.0	8.0	7.6	8.1	
	5% WB	MCDBR	13.7	14.7	17.4	20.0	18.0	17.1	16.8	15.9	16.4	14.0	14.5	13.0	
		MCWBR	8.0	8.0	9.0	9.6	7.9	7.0	6.8	5.4	6.5	7.2	7.6	8.0	
Clear-Sky Solar Irradiance	taub	0.364	0.392	0.428	0.468	0.434	0.423	0.427	0.434	0.423	0.404	0.374	0.365		
	taud	2.343	2.247	2.158	2.064	2.238	2.312	2.314	2.293	2.308	2.333	2.361	2.346		
	Ebn at Noon	245	255	259	256	269	271	269	263	258	248	239	233		
	Edh at Noon	28	36	43	51	44	41	40	40	37	32	27	26		
All-Sky Solar Radiation	RadAvg	478	698	1103	1529	1935	2194	2257	1957	1455	953	629	442		
	RadStd	71	102	105	169	159	179	81	114	111	85	83	67		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	-176	N/A	+79

Nomenclature: See separate page